

Altair Amplifier Board Layout

Every part of the 8-channel amplifier board, hole by hole, on a full-size breadboard layout (rows numbered 1–60, columns a–j, two power rails on each side). Build it on a solderless 830-point breadboard to test, then solder the identical layout onto an Adafruit Perma-Proto Full-sized board. Sheet 0 shows how the ESP32 and the other ready-made boards connect to it, with every wire numbered.

HOW TO READ IT

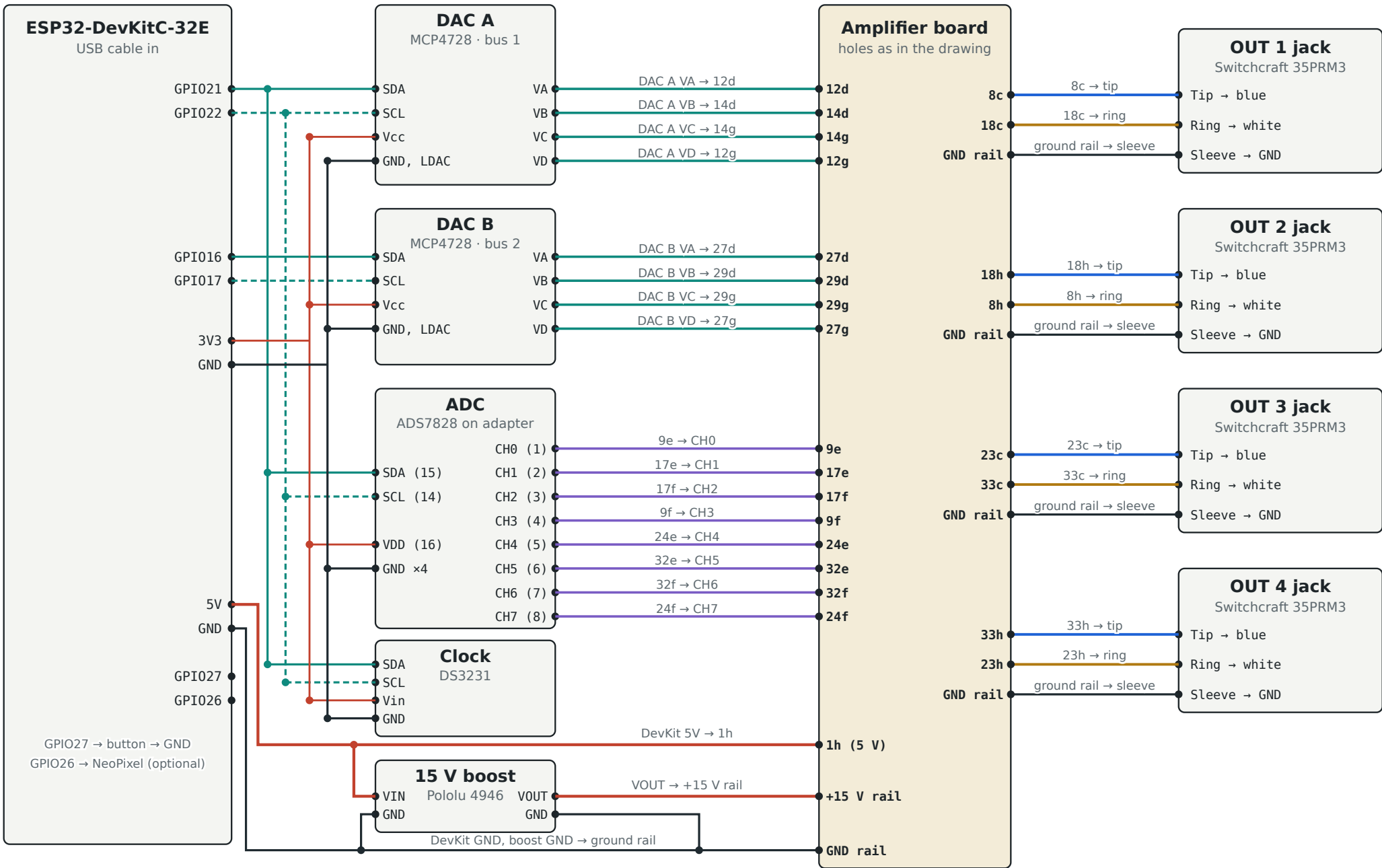
1. Each **row** has two strips of 5 connected holes: **a-e** on the left, **f-j** on the right. Anything in the same strip is connected. The chips straddle the middle.
2. The outer rails on both sides are **+15 V** (red), the inner rails are **ground** (black).
3. A part is drawn as a line between the two holes its legs go in. Lines that cross are fine: lift one part a little.
4. Coloured dots are wires that leave the board: teal to the DACs, blue to the jacks, purple to the ADC, red to 5 V.
5. The orange dashed wires carry –5 V from the TC7660 to pin 11 of each LM2902.

 resistor  ceramic capacitor  electrolytic (mind + / –)  TVS diode  BAT85 (stripe toward the strip)

Labels on the parts: 1k = 1 kΩ ½ W, 39k = 39 kΩ, 10k = 10 kΩ, 60k4 = 60.4 kΩ, 1n = 1 nF C0G, 10n = 10 nF, 100n = 100 nF, TVS = P6KE15CA, D = BAT85, 10μ / 470μ = electrolytics. Dot labels: 1T = OUT 1 tip, 1R = OUT 1 ring, A·VA = DAC A output VA, CH0 = ADC channel 0.

The ESP32 and the other boards

The amplifier board is the only one you build from parts. The ESP32, the two DACs, the ADC, the clock and the 15 V boost are ready-made boards that sit beside it in the box, joined by wires. Every wire is drawn on its own. Each label says where the wire lands: "12d" is the hole in row 12, column d of the amplifier board drawing. Solid teal is SDA, dashed teal is SCL, red is power, black is ground.



Every wire, numbered

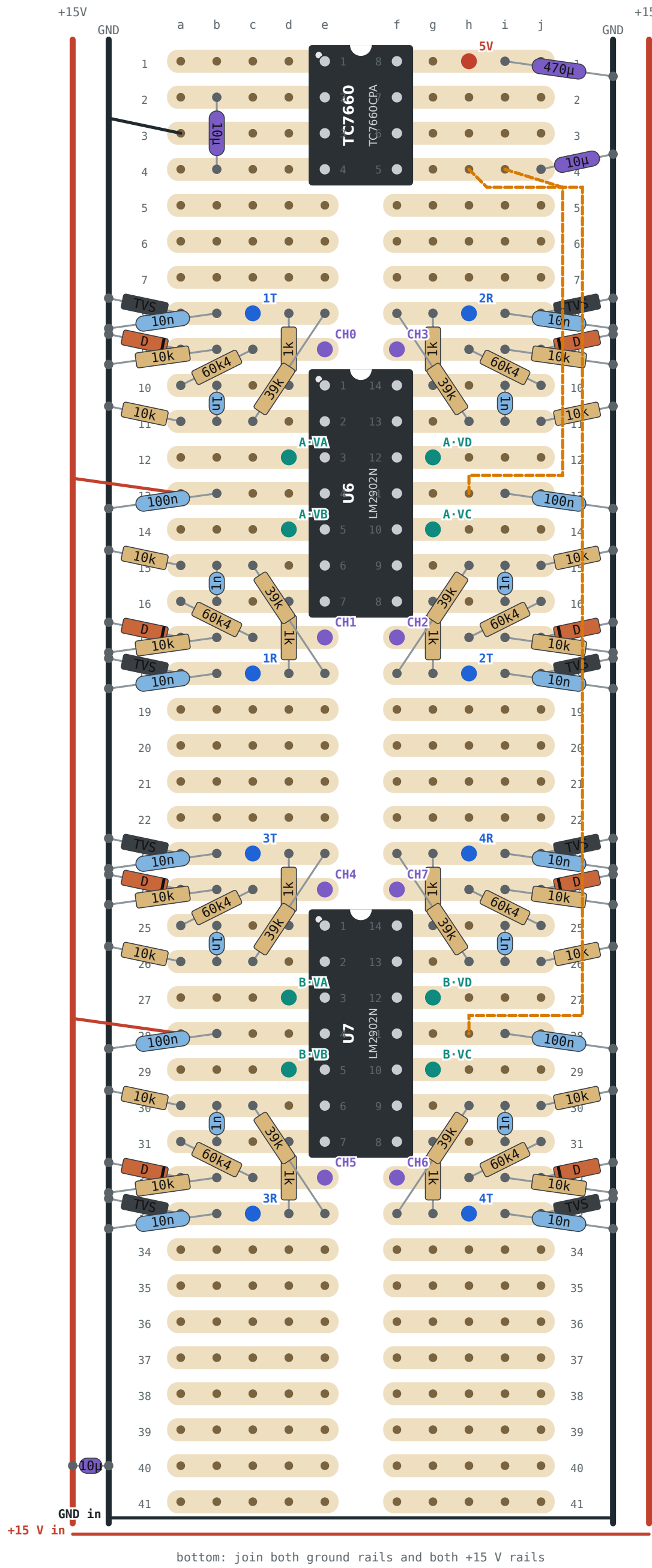
"Amp board 12d" means the hole in row 12, column d of the amplifier board drawing below.

#	FROM	TO	NOTES
Power			
1	DevKit 5V	Amp board 1h	5 V to the TC7660
2	DevKit 5V	Pololu VIN	
3	DevKit GND	Amp board ground rail (black)	
4	Pololu GND	Amp board ground rail (black)	
5	Pololu VOUT	Amp board +15 V rail (red)	
3.3 V and ground to the small boards			
6	DevKit 3V3	DAC A Vcc	
7	DevKit 3V3	DAC B Vcc	
8	DevKit 3V3	RTC Vin	
9	DevKit 3V3	ADC pin 16 (+VDD)	
10	DevKit GND	DAC A GND	
11	DevKit GND	DAC A LDAC	
12	DevKit GND	DAC B GND	
13	DevKit GND	DAC B LDAC	
14	DevKit GND	RTC GND	
15	DevKit GND	ADC pin 9 (GND)	
16	DevKit GND	ADC pin 11 (COM)	
17	DevKit GND	ADC pin 12 (A0)	
18	DevKit GND	ADC pin 13 (A1)	
19	ADC pin 10 (REF)	2.2 μF capacitor to GND	on the ADC adapter

#	FROM	TO	NOTES
20	ADC pin 16 (+VDD)	100 nF capacitor to GND	on the ADC adapter
I²C bus 1 (data wires)			
21	DevKit GPIO21	DAC A SDA	
22	DevKit GPIO21	ADC pin 15 (SDA)	
23	DevKit GPIO21	RTC SDA	
24	DevKit GPIO22	DAC A SCL	
25	DevKit GPIO22	ADC pin 14 (SCL)	
26	DevKit GPIO22	RTC SCL	
I²C bus 2 (DAC B only)			
27	DevKit GPIO16	DAC B SDA	
28	DevKit GPIO17	DAC B SCL	
DAC outputs to the amplifier board			
29	DAC A VA	Amp board 12d	
30	DAC A VB	Amp board 14d	
31	DAC A VC	Amp board 14g	
32	DAC A VD	Amp board 12g	
33	DAC B VA	Amp board 27d	
34	DAC B VB	Amp board 29d	
35	DAC B VC	Amp board 29g	
36	DAC B VD	Amp board 27g	
Amplifier board to the ADC (monitor)			
37	Amp board 9e	ADC pin 1 (CH0)	
38	Amp board 17e	ADC pin 2 (CH1)	
39	Amp board 17f	ADC pin 3 (CH2)	
40	Amp board 9f	ADC pin 4 (CH3)	
41	Amp board 24e	ADC pin 5 (CH4)	
42	Amp board 32e	ADC pin 6 (CH5)	
43	Amp board 32f	ADC pin 7 (CH6)	
44	Amp board 24f	ADC pin 8 (CH7)	
Amplifier board to the jacks			
45	Amp board 8c	OUT 1 jack, tip lug	
46	Amp board 18c	OUT 1 jack, ring lug	
47	Amp board ground rail (black)	OUT 1 jack, sleeve lug	
48	Amp board 18h	OUT 2 jack, tip lug	
49	Amp board 8h	OUT 2 jack, ring lug	
50	Amp board ground rail (black)	OUT 2 jack, sleeve lug	
51	Amp board 23c	OUT 3 jack, tip lug	
52	Amp board 33c	OUT 3 jack, ring lug	
53	Amp board ground rail (black)	OUT 3 jack, sleeve lug	
54	Amp board 33h	OUT 4 jack, tip lug	
55	Amp board 23h	OUT 4 jack, ring lug	
56	Amp board ground rail (black)	OUT 4 jack, sleeve lug	
Button and status light			
57	DevKit GPIO27	Push button, one side	
58	Push button, other side	DevKit GND	
59	DevKit GPIO26	NeoPixel DIN	optional
60	DevKit 5V	NeoPixel 5V / VIN	optional
61	DevKit GND	NeoPixel GND	optional

Rows 1-41

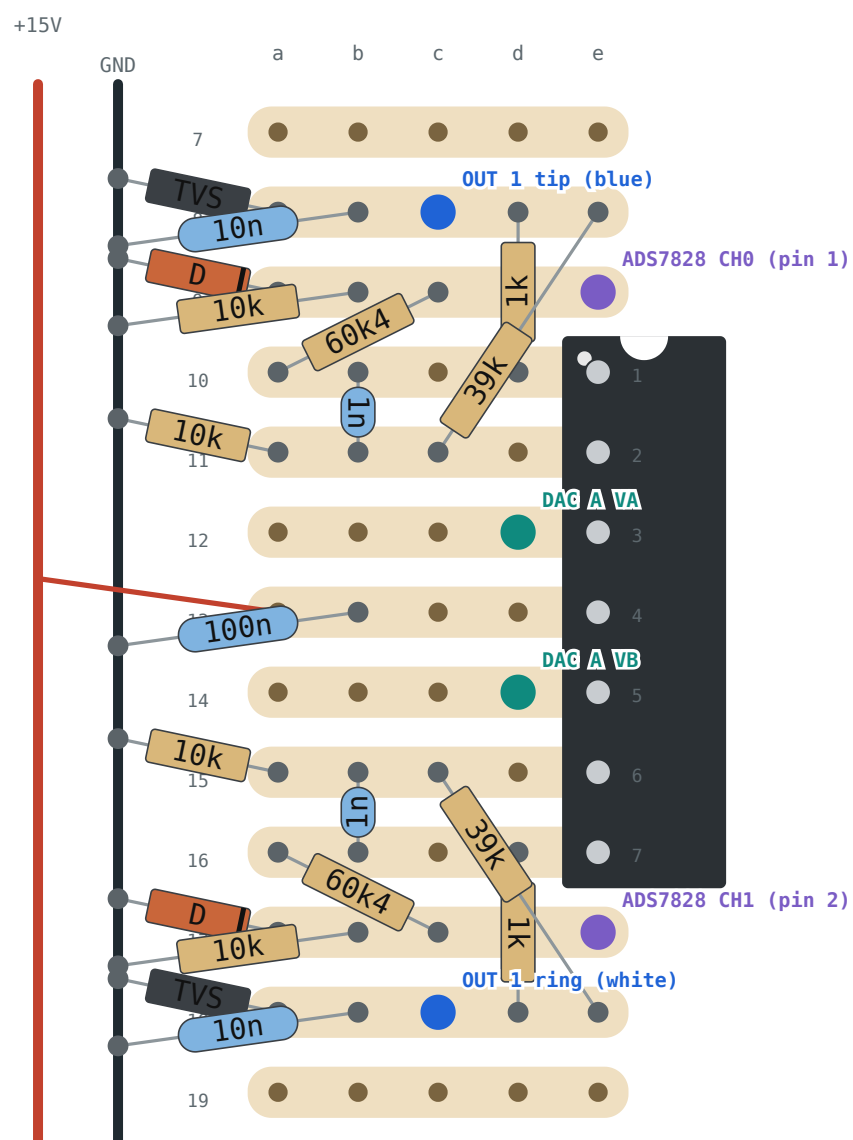
U6 drives OUT 1 and OUT 2, U7 drives OUT 3 and OUT 4. Rows 42-60 stay empty.



At the bottom of the board: wire the left +15 V rail to the right +15 V rail, and the left ground rail to the right ground rail. Connect +15 V from the Pololu VOUT to a red rail, and ground (DevKit GND, Pololu GND, all 4 jack sleeves) to a black rail.

U6 amp A and amp B (the left side of U6)

All 8 channels use these same 9 parts. The right side of each chip is the mirror image, and U7 is a copy of U6 fifteen rows lower.



The 39 kΩ goes from the NIN row (row 11, pin 2) to the **jack row** (row 8), the same strip as the far end of the 1 kΩ. That is what keeps the voltage exact with 6 drivers chained.

Placement list

Hole names are row then column: 10d is row 10, column d. Tick them off as you go.

PART	LEG 1	LEG 2	NOTES
−5 V supply (TC7660, rows 1-4)			
10 μF	2b	4b	+ lead in row 2 (pin 2), − lead in row 4 (pin 4)
wire	3a	left ground rail	pin 3 to ground
10 μF	4j	right ground rail	− lead in row 4 (pin 5, −5 V), + lead on the ground rail
wire	4i	13h	−5 V to U6 pin 11
wire	4h	28h	−5 V to U7 pin 11
470 μF 16 V	1i	right ground rail	+ lead in row 1 (5 V), − lead on the ground rail
U6 power pins			
wire	13a	left +15 V rail	pin 4 to +15 V
100 nF	13b	left ground rail	pin 4 decoupling
100 nF	13i	right ground rail	pin 11 decoupling
U6 amp A → OUT 1 blue			
1 nF C0G	10b	11b	stability
10 kΩ	11a	left ground rail	gain
1 kΩ ½ W	10d	8d	output, short protection
39 kΩ	11c	8e	feedback (jack side)
60.4 kΩ	10a	9c	monitor divider
P6KE15CA	8a	left ground rail	either way round
10 nF	8b	left ground rail	noise filter
BAT85	9a	left ground rail	stripe on the row side
10 kΩ	9b	left ground rail	monitor divider
U6 amp B → OUT 1 white			
1 nF C0G	16b	15b	stability
10 kΩ	15a	left ground rail	gain
1 kΩ ½ W	16d	18d	output, short protection
39 kΩ	15c	18e	feedback (jack side)
60.4 kΩ	16a	17c	monitor divider
P6KE15CA	18a	left ground rail	either way round
10 nF	18b	left ground rail	noise filter
BAT85	17a	left ground rail	stripe on the row side
10 kΩ	17b	left ground rail	monitor divider
U6 amp C → OUT 2 blue			
1 nF C0G	16i	15i	stability
10 kΩ	15j	right ground rail	gain
1 kΩ ½ W	16g	18g	output, short protection
39 kΩ	15h	18f	feedback (jack side)
60.4 kΩ	16j	17h	monitor divider
P6KE15CA	18j	right ground rail	either way round
10 nF	18i	right ground rail	noise filter
BAT85	17j	right ground rail	stripe on the row side
10 kΩ	17i	right ground rail	monitor divider
U6 amp D → OUT 2 white			
1 nF C0G	10i	11i	stability
10 kΩ	11j	right ground rail	gain
1 kΩ ½ W	10g	8g	output, short protection
39 kΩ	11h	8f	feedback (jack side)
60.4 kΩ	10j	9h	monitor divider
P6KE15CA	8j	right ground rail	either way round
10 nF	8i	right ground rail	noise filter

PART	LEG 1	LEG 2	NOTES
BAT85	9j	right ground rail	stripe on the row side
10 kΩ	9i	right ground rail	monitor divider
U7 power pins			
wire	28a	left +15 V rail	pin 4 to +15 V
100 nF	28b	left ground rail	pin 4 decoupling
100 nF	28i	right ground rail	pin 11 decoupling
U7 amp A → OUT 3 blue			
1 nF C0G	25b	26b	stability
10 kΩ	26a	left ground rail	gain
1 kΩ ½ W	25d	23d	output, short protection
39 kΩ	26c	23e	feedback (jack side)
60.4 kΩ	25a	24c	monitor divider
P6KE15CA	23a	left ground rail	either way round
10 nF	23b	left ground rail	noise filter
BAT85	24a	left ground rail	stripe on the row side
10 kΩ	24b	left ground rail	monitor divider
U7 amp B → OUT 3 white			
1 nF C0G	31b	30b	stability
10 kΩ	30a	left ground rail	gain
1 kΩ ½ W	31d	33d	output, short protection
39 kΩ	30c	33e	feedback (jack side)
60.4 kΩ	31a	32c	monitor divider
P6KE15CA	33a	left ground rail	either way round
10 nF	33b	left ground rail	noise filter
BAT85	32a	left ground rail	stripe on the row side
10 kΩ	32b	left ground rail	monitor divider
U7 amp C → OUT 4 blue			
1 nF C0G	31i	30i	stability
10 kΩ	30j	right ground rail	gain
1 kΩ ½ W	31g	33g	output, short protection
39 kΩ	30h	33f	feedback (jack side)
60.4 kΩ	31j	32h	monitor divider
P6KE15CA	33j	right ground rail	either way round
10 nF	33i	right ground rail	noise filter
BAT85	32j	right ground rail	stripe on the row side
10 kΩ	32i	right ground rail	monitor divider
U7 amp D → OUT 4 white			
1 nF C0G	25i	26i	stability
10 kΩ	26j	right ground rail	gain
1 kΩ ½ W	25g	23g	output, short protection
39 kΩ	26h	23f	feedback (jack side)
60.4 kΩ	25j	24h	monitor divider
P6KE15CA	23j	right ground rail	either way round
10 nF	23i	right ground rail	noise filter
BAT85	24j	right ground rail	stripe on the row side
10 kΩ	24i	right ground rail	monitor divider
Rails			
wire	left +15 V rail	right +15 V rail	join the two +15 V rails (bottom of board)
wire	left ground rail	right ground rail	join the two ground rails (bottom of board)
10 μF	left +15 V rail (row 40)	left ground rail (row 40)	across the left rails, + on +15 V

Connections to the other boards and the jacks

LABEL ON MAP	HOLE	WIRE TO
5V	1h	5 V from DevKit 5V pin
A·VA	12d	DAC A VA
A·VB	14d	DAC A VB
A·VC	14g	DAC A VC
A·VD	12g	DAC A VD
B·VA	27d	DAC B VA
B·VB	29d	DAC B VB
B·VC	29g	DAC B VC
B·VD	27g	DAC B VD
1R	18c	OUT 1 ring (white)
1T	8c	OUT 1 tip (blue)
2R	8h	OUT 2 ring (white)
2T	18h	OUT 2 tip (blue)
3R	33c	OUT 3 ring (white)
3T	23c	OUT 3 tip (blue)
4R	23h	OUT 4 ring (white)
4T	33h	OUT 4 tip (blue)
CH0	9e	ADS7828 CH0 (pin 1)
CH1	17e	ADS7828 CH1 (pin 2)
CH2	17f	ADS7828 CH2 (pin 3)
CH3	9f	ADS7828 CH3 (pin 4)
CH4	24e	ADS7828 CH4 (pin 5)
CH5	32e	ADS7828 CH5 (pin 6)
CH6	32f	ADS7828 CH6 (pin 7)
CH7	24f	ADS7828 CH7 (pin 8)

The DevKit, DAC, ADC, clock and boost connections to each other are in the wiring PDF (table 2).

BUILD ORDER

Test as you go

1. Rails and the TC7660 with its two capacitors. Power up: row 4 should read about −4.8 V.
2. U6 with its power wires and two 100 nF. Then the 9 parts of amp A. Connect DAC A and the DevKit, flash firmware 2.2.2, set Manual 100 %: row 8 reads 10 V (before calibration it may read 9.8–10.2 V).
3. The other 7 channels, testing each one the same way.
4. Jack wires and the ADC wires last.